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**Summer Term 2026**

# Pattern Matching and Machine Learning for Audio Signal Processing
# Exercise sheet 1

To be uploaded in eCampus till: 24-04-2026 22:00 (strict deadline)

The solutions of the exercises have to be handed in via eCampus (you have to upload them) till the date and time reported on top of each exercise sheet. For all the other information about the exercises please refer to the eCampus page of the course.

## Exercise 1.1

[$2+2+2+2=8$ points]

(a) Calculate (without using a calculator) the polar coordinates representation of the following complex number:
$$z = \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)^3.$$

(b) Write down the following complex number in Cartesian coordinates:
$$z = \frac{e^{\frac{5\pi}{3}i}}{e^{\frac{4}{3}\pi i} \cdot 2e^{\frac{11\pi}{6}i}}.$$

(c) Calculate explicitly $\mathrm{Re}\left[5e^{\frac{\pi}{4}i} + \sqrt{2}e^{\pi i}\right]$, where $\mathrm{Re}[z]$ indicates the real part of a complex number $z$.

(d) Using the Euler formula ($e^{iz} = \cos(z) + i\sin(x)$), prove the following statement:
$\sin(z)^2 + \cos(z)^2 = 1, \qquad z \in \mathbb{C}.$

## Exercise 1.2

[$2+2+2+1=7$ points]

In this exercise you will have to solve small programming tasks. Please follow the rules for programming tasks written in eCampus.

(a) Create a 5 seconds long discrete sine wave of frequency 1000 Hz.The sampling rate is 44100 Hz.

(b) In the introductory slides you have seen that you can illustrate a signal using a time-frequency representation, the windowed Fourier Transform. This can be done in Matlab using the command `spectrogram` (type in Matlab `help spectrogram` to have more information). Plot the spectrogram of the signal you created in (a).

(c) Repeat (a) and (b) with a cosine wave of frequency 1000 Hz. Compare the two spectrograms. What can you see?

(d) Save the signal you created in (a) in a .wav file.